Problem 9

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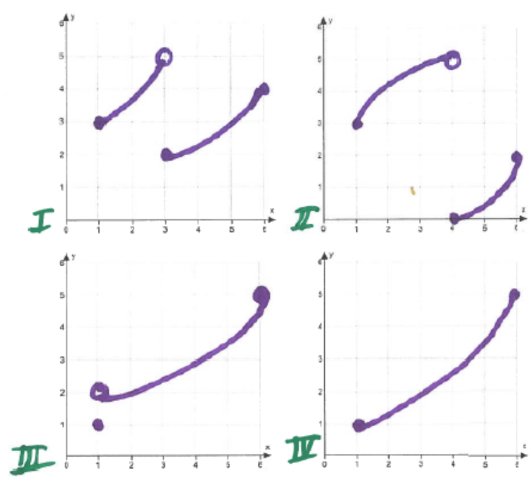
Problem 1 (a) *Given function f on an interval $[a, b]$:*

(i) *What are the conditions of the Intermediate Value Theorem?*

(ii) *What is the conclusion of the Intermediate Value Theorem?*

(b) *Given the four functions on the interval $[1, 6]$, answer the questions below.*



(i) *For each of the functions I through IV, indicate $f(1)$ and $f(6)$. Then mark the interval of all numbers strictly between $f(1)$ and $f(6)$, on the y-axis.*

(ii) *For each of the functions I through IV, write an interval of all numbers strictly between $f(1)$ and $f(6)$*

(iii) *List the functions that satisfy the conditions of the Intermediate Value Theorem on $[1, 6]$*

(iv) *For which of the functions is the following statement true: For any number L strictly between $f(1)$ and $f(6)$, there exists a number c in $(1, 6)$ satisfying $f(c) = L$.*

(v) *Does the function III satisfy the conclusion of the Intermediate Value Theorem? Why or why not?*